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China Silver Futures Research Framework 中国白银期货研究框架

2025/12/12

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China Futures Research Framework

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CONTENTS

Industrial Chain

供需概况

01

Silver Contract

白银合约

02

Pricing Model

定价模式

03



Silver supply mainly from mineral production and recycling; demand consists of industrial use, investment use, photographic use, and jewelry use. Regarding silver prices: first, the silver inventory-to-demand ratio exceeds 100%, resulting in limited impact of changes in supply-demand gaps on prices; second, silver prices mainly fluctuate following gold prices, with elasticity influenced by industrial attributes; third, the gold-silver ratio is negatively correlated with copper prices and rises during the Federal Reserve's interest rate cutting cycle.

China's silver contracts include physical trading and margin trading (Au (T+D)) on the Shanghai Gold Exchange (SGE), as well as the silver futures contract (Ag) on the Shanghai Futures Exchange (SHFE). Major overseas silver contracts consist of London Silver Spot and COMEX Silver Futures. When calculating the China-overseas price spread, attention should be paid to differences in measurement units, exchange rate fluctuations, and value-added tax (VAT).

白银供应主要来自矿产开发和回收利用，需求分为工业用银、投资用银、摄影用银以及首饰用银。银价方面，一是，白银库存需求比超过100%，导致供需缺口变动对价格的影响较小；二是，银价主要跟随金价波动，弹性受工业属性影响；三是，金银比价和铜价负相关，并在美联储降息周期中上行。

中国白银合约有上海黄金交易所的Ag99.99现货即期合约、Ag(T+D)的现货延期交收合约，以及上海期货交易所的Ag白银期货合约。外盘白银合约主要有伦敦银现和COMEX白银期货。计算内外价差时，需要注意计量单位、汇率波动和增值税差异。

Risks: Monetary policy, geopolitics, and rule revisions, Etc.

风险提示：货币政策、地缘政治、规则修订等



Industrial Chain

产业链

01

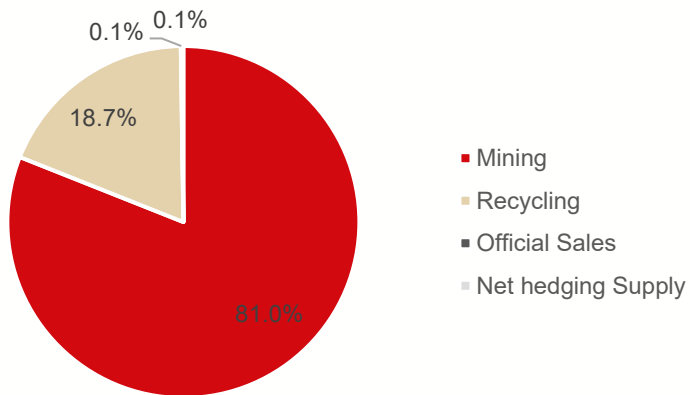
1.1 Industry Chain

产业链

- Supply: mining, recycling, official sales, etc.
- Demand: Primarily comprises industrial use, investment demand, photographic applications, and jewelry manufacturing.
- 供给：分为矿产、回收利用、官方销售。
- 需求：主要由工业用银、投资用银、摄影用银以及首饰用银组成。

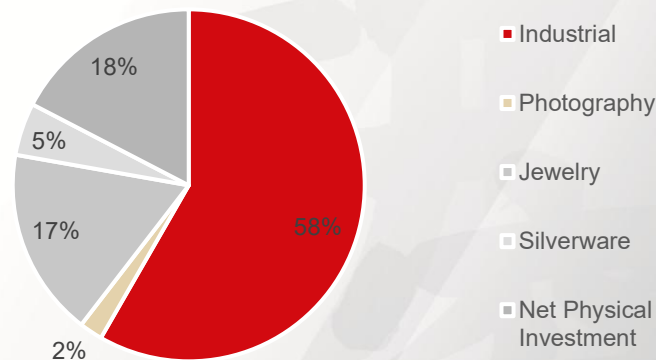
Silver Supply Breakdown %

白银供给构成



Silver Demand Breakdown %

白银需求构成



Sources: Silver Institute, CITIC Futures

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1.2.1 Balance Sheet: Global

全球供需平衡表

Table: Global Silver Supply and Demand (Million ounces)

Supply	2021	2022	2023	2024	2025F
Mine Production	831	839	813	820	835
Recycling	191	194	184	194	193
Net Hedging Supply	-	-	-	-	1
Net Official Sector Sales	2	2	2	2	2
Total Supply	1,023	1,035	998	1,015	1,031
Demand	2021	2022	2023	2024	2025F
Industrial (total)	564	592	657	681	677
Electrical & Electronics	351	371	444	461	466
...of which photovoltaics	89	118	193	198	196
Brazing Alloys & Solders	51	49	50	52	53
Other Industrial	163	172	163	168	159
Photography	28	28	27	26	24
Jewelry	182	235	203	209	196
Silverware	41	74	55	54	46
Coin & Net Bar Demand	284	338	244	191	204
Net Hedging Demand	4	18	12	4	0
Total Demand	1103	1284	1199	1164	1148
Total Demand ex Net Physical Investment	819	946	955	973	944
Market Balance	-80	-250	-201	-149	-118
Net Investment in ETPs	65	-117	-38	62	70
Market Balance less ETPs	-145	-132	-164	-210	-188
Industrial Balance (Total Supply – Total Demand ex Net Physical Investment)	205	89	43	42	87
Silver Price (US\$/oz, London price)	25	22	23	28	0

表：全球白银供需平衡表（单位：百万盎司）

供给	2021	2022	2023	2024	2025F
矿产	831	839	813	820	835
回收	191	194	184	194	193
对冲净供应	-	-	-	-	1
政府净出售	2	2	2	2	2
总供给	1,023	1,035	998	1,015	1,031
需求	2021	2022	2023	2024	2025F
工业	564	592	657	681	677
电子电器	351	371	444	461	466
其中：光伏	89	118	193	198	196
钎料与焊料	51	49	50	52	53
其他工业	163	172	163	168	159
摄影	28	28	27	26	24
银饰	182	235	203	209	196
银器	41	74	55	54	46
净实物投资	284	338	244	191	204
净对冲净需求	4	18	12	4	0
总需求	1103	1284	1199	1164	1148
总需求EX净实物投资	819	946	955	973	944
总平衡（供给-需求）	-80	-250	-201	-149	-118
净ETP投资	65	-117	-38	62	70
剔除ETP后的市场平衡	-144	-132	-163	-211	-188
工业平衡（总供给-总需求EX净实物投资）	205	89	44	42	87
银价	25	22	23	28	0

Sources: Silver Institute, CITIC Futures

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1.2.2 Balance Sheet: China

中国供需平衡表

Table: China Silver Supply and Demand (Million ounces)

Supply	2021	2022	2023	2024
Mine Production	113	112	112	110
Recycling	33	37	37	40
Total Supply	146	149	149	150
Demand	2021	2022	2023	2024
Industrial (total)	152	184	257	275
Electrical & Electronics	90	124	196	211
Brazing Alloys & Solders	22	20	20	21
Other Industrial	40	40	41	43
Jewelry	21	17	16	15
Silverware	3	2	2	2
Physical Investment	8	7	6	6
Total Demand	184	211	282	298
Total Demand ex Net Physical Investment	176	203	276	293
Total Supply	146	149	149	150
Total Demand	184	211	282	298
Total Demand ex Net Physical Investment	176	203	276	293
Market Balance	-38	-61	-133	-148
Industrial Balance	-30	-54	-127	-143
Silver Price (US\$/oz, London price)	25	22	23	28

表：中国白银供给与需求

供给	2021	2022	2023	2024
矿产	113	111.8	112	110
回收	33	37	37	40
总供给	146	149	149	150
需求	2021	2022	2023	2024
工业(总)	152	184	257	275
电子电器	90	124	196	211
钎料与焊料	22	20	20	21
其他工业	40	40	41	43
银饰	21	17	16	15
银器	3	2	2	2
净实物投资	8	7	6	6
总需求	184	211	282	298
总需求EX净实物投资	176	203	276	293
总供给	146	149	149	150
总需求	184	211	282	298
总需求EX净实物投资	176	203	276	293
总平衡（供给-需求）	-38	-61	-133	-148
工业平衡	-30	-54	-127	-143
银价	25	22	23	28

Sources: Silver Institute, CITIC Futures

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Silver Contract

白银合约

02

2.1 SGE Margin Contracts

上金所白银延期合约

合约	上金所白银延期合约
合约代码	Ag(T+D)
交易方式	现货延期交收交易
交易单位	1千克/手
报价单位	元（人民币）/千克
最小变动价位	1 元/千克
最小单笔报价	1 手
最大单笔报价	10000手
合约期限	连续交易
交易时间	日间：9:00至15:30，夜间:19:50 至次日 02:30
延期补偿费收付日	按自然日逐日收付
最小交收申报量	1手
交收申报时间	15:00—15:30
中立申报时间	15:31—15:40
清算方式	当日无负债清算制度
实物交收方式	交收申报制
交割方式	实物交割
交割品种	标准重量 15 千克、成色不低于 99.99%的银锭
交割时间	交收申报配对成功的当日
交割地点	交易所指定仓库

Product	Silver
Contract Symbol	Ag(T+D)
Trading Market	Matching
Trading Unit	1 kg/lot
Quotation Unit	yuan/kg
Minimum Price Fluctuation	1 Yuan/gram
Minimum Quotation Size	1 lot
Maximum Quotation Size	10000 lot
Contract Term	Continuous Trading
Trading Hours	Day Session: 9:00 -15:30; Night Session: 20:00 -02:30
Settlement Date of Deferred Compensation	Daily Settlement on Natural Days
Minimum Settlement Declaration Quantity	1 lot
Settlement Declaration Time Window	15:00—15:30
Neutral Declaration Time Window	15:31—15:40
Settlement method	Daily mark-to-market Settlement
Delivery Mode	Physical delivery
Deliverable Bullions	Silver ingots with a standard weight of 15 kg and a fineness not less than 99.99%
Time of Delivery	The same day as delivery requests are successfully matched
Delivery Location	SGE Certified Vault

2.2 SHFE Futures Contract

上期所白银期货合约

合约	上期所白银期货合约
交易单位	15千克/手
报价单位	元（人民币）/克
最小变动价位	1元（人民币）/克
涨跌停板幅度	上一交易日结算价±3%
合约月份	1-12月
交易时间	上午9:00—11:30，下午1:30—3:00和交易所规定的其他交易时间
最后交易日	标合约月份的15日（遇国家法定节假日顺延，春节月份等最后交易日交易所可另行调整并通知）
交割日期	最后交易日后第连续二个工作日
交割品级	标准品：符合国标GB/T 4135-2016 IC-Ag99.99规定，其中银含量不低于99.99%。
交割地点	交易所交割仓库
最低交易保证金	合约价值的4%
交割方式	实物交割
交割单位	30千克
交易代码	AG

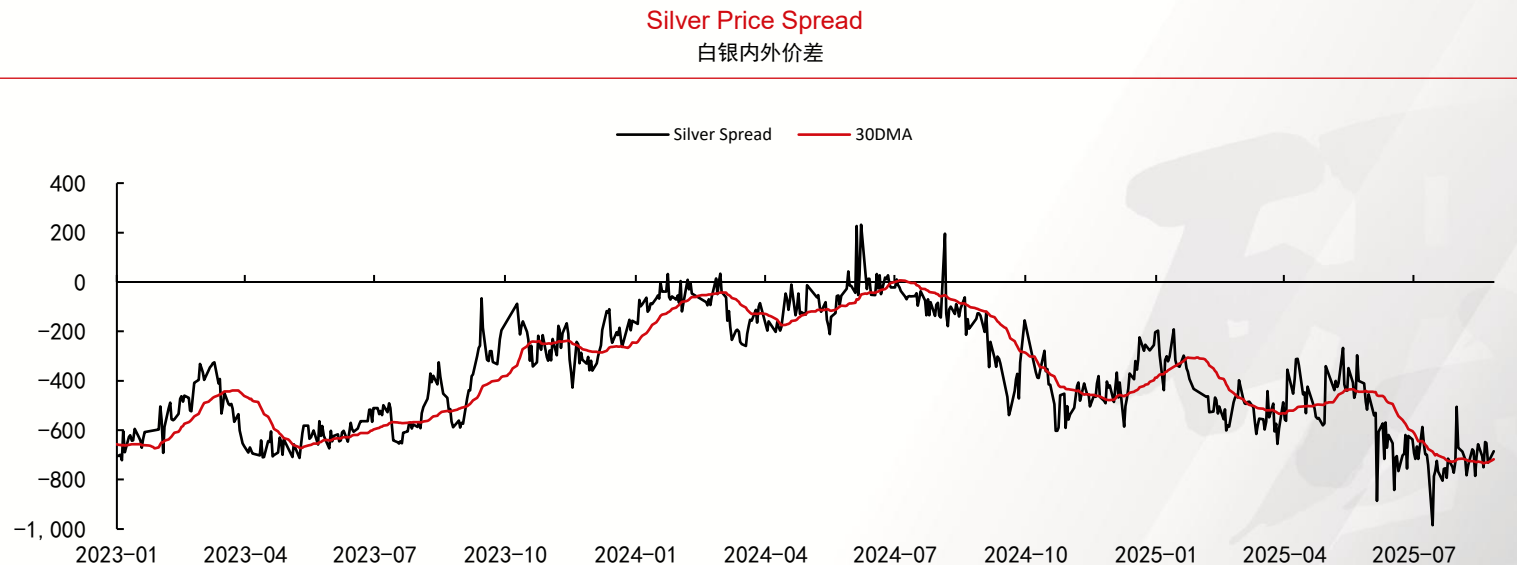
Product	Silver
Contract Size	15 kilograms/lot
Price Quotation	Yuan (RMB)/gram
Minimum Price Fluctuation	1 yuan/gram
Range of Price Limit	Within 3% of the settlement price of the preceding trading day
Listed Contracts	Months 1 through 12
Trading Hours	9:00 a.m. to 11:30 a.m., 1:30 p.m. to 3:00 p.m., and other hours specified by the Exchange (Beijing Time)
Last Trading Day	15th day of the contract month (postponed accordingly if it is a legal holiday in China and subject to separate adjustment and announcement by the Exchange if it falls in the Spring Festival month or another month specially designated by the Exchange.)
Delivery Period	Two consecutive business days starting from the day after the last trading day
Grade and Quality Specifications	Standard Article: Meeting GB/T 4135-2016 (IC-Ag99.99) – Silver Content ≥ 99.99%.
Delivery Venue	SHFE-designated delivery warehouses
Minimum Trade Margin	4% of contract value
Settlement Type	Physical delivery
Delivery Unit	30 kilograms
Contract Symbol	AG

Sources: SHFE, CITIC Futures

2.3 Silver Price Spread

白银内外价差

- Price Spread = SGE Silver – Loco London/31.1035 (troy ounce) * VAT rate * RMB exchange rate
- 内外价差 = 内盘人民币计价白银 - 外盘美元计价白银/31.1035(金衡盎司) * 增值税率 * 人民币汇率



Sources: Wind, CITIC Futures



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Pricing Model

定价模式

03

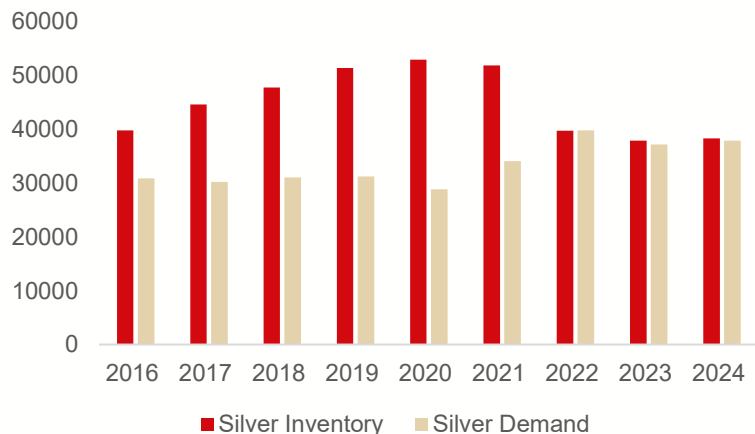
3.1 Limited Sensitivity to Supply-Demand Fluctuations

银价受供需影响较小

- With silver's inventory-to-demand ratio exceeding 100% and its absolute inventory level being relatively high, fluctuations in the supply-demand gap have a diminished impact on prices.
- 白银库存需求比超过100%，库存绝对水平较高，导致供需缺口变动对价格的影响较小。

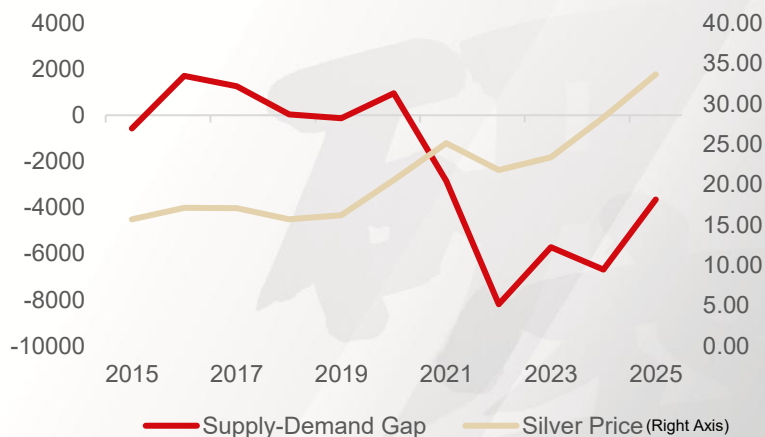
Silver Inventory & Demand (ton)

白银库存和需求



Silver Price & Supply-Demand GAP (ton, \$)

白银价格与供需缺口



Sources: Wind, Silver Institute, CITIC Futures

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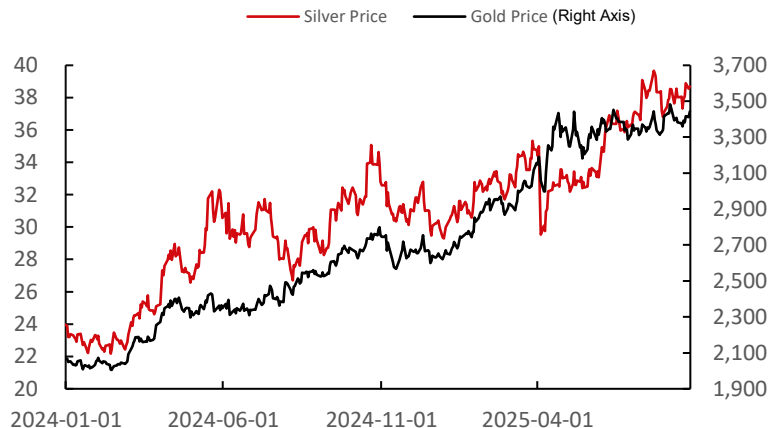
3.2 Gold Sets the Trend, with Industrial Demand Shaping Price Sensitivity

黄金主导趋势，工业属性影响弹性

- Silver is inherently linked to gold, with its absolute valuation and price trends predominantly following those of gold. The gold-silver ratio serves as an indicator of relative elasticity, exhibiting a negative correlation with copper prices—a dynamic primarily reflecting shifts in industrial demand characteristics, which are largely driven by macroeconomic demand cycles.
- 白银绝对价格跟随黄金。金银比表征两者相对弹性，和铜价呈负相关关系，主要由于白银工业属性受宏观需求周期影响。

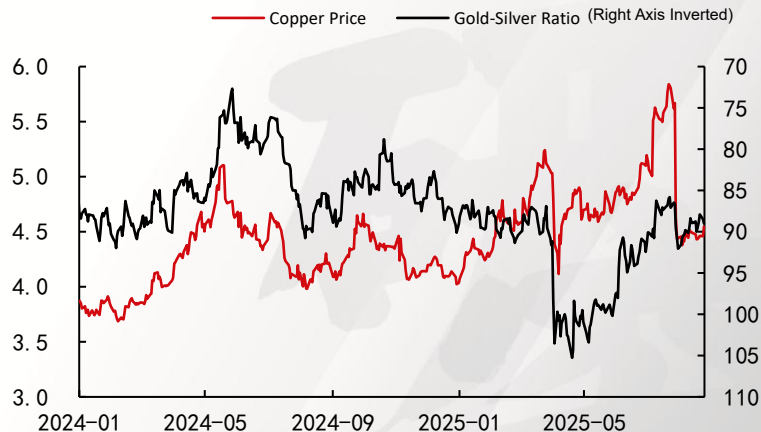
Gold and Silver Price

黄金和白银价格



Copper Price and Gold-silver Ratio

铜价和金银比值



Sources: Wind, CITIC Futures

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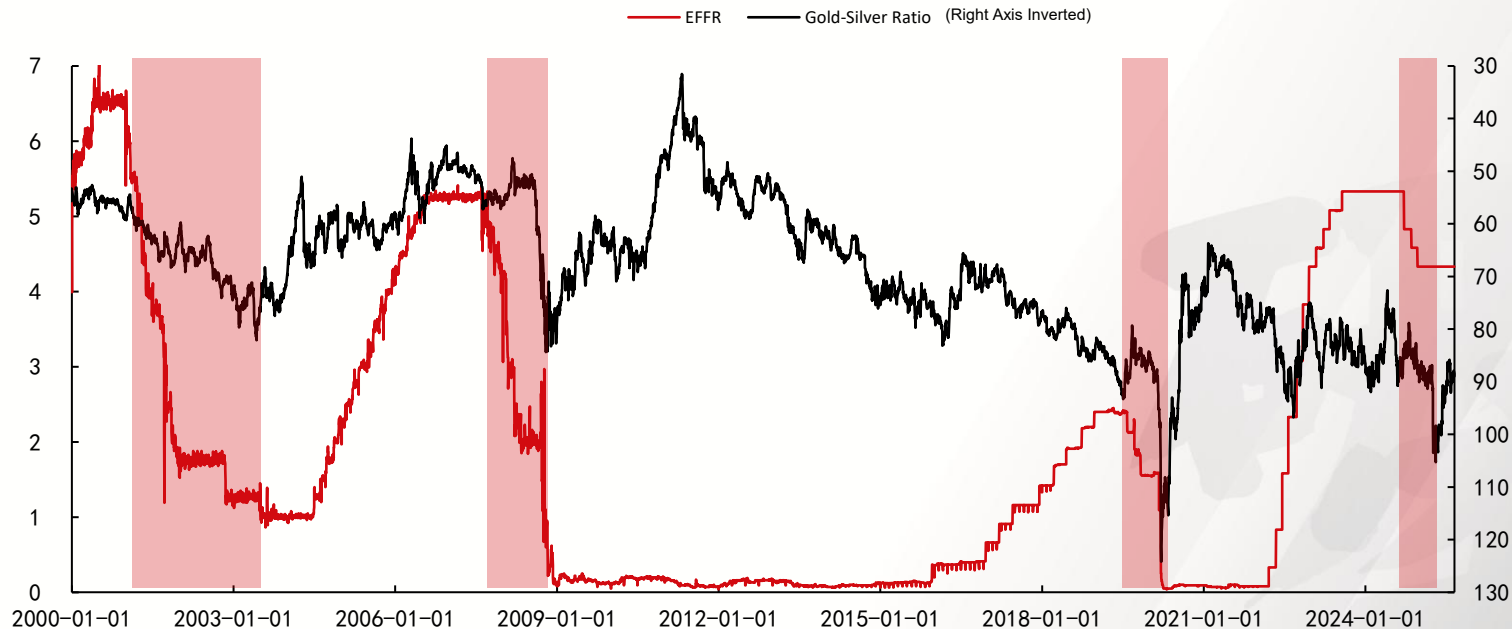
3.3 Gold-Silver Ratio and Monetary Policy Cycle

金银比值和货币周期



During the Federal Reserve's rate-cutting cycle, the gold-silver ratio rises.

美联储降息周期中金银比值上行



Sources: Wind, CITIC Futures



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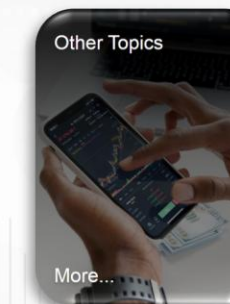
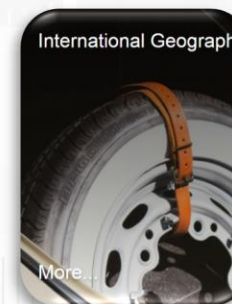
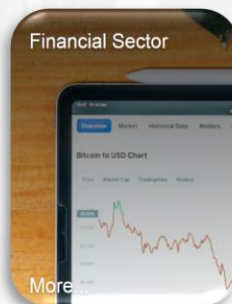
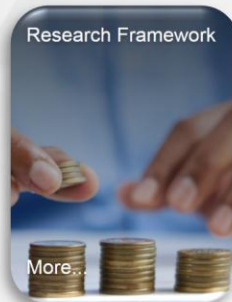
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